



Research Article

Effect of luminance of head-up displays on recognition of visual objects on roads

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ABSTRACT

Objective: The purpose of this study is to investigate the effect of visual obstruction when a head-up display (HUD) is presented in front of a driver's field of view.

Background: The use of HUDs is expected to increase due to advances in augmented reality (AR) technology. A conventional HUD is displayed at a relatively low position in relation to the driver's viewpoint and has little effect on their front view. However, while an HUD, such as the AR-HUD, is presented in front of the driver's field of vision and helps them in presenting ADAS information, there is a concern that this display may obscure visual objects on the road from the driver's field of vision.

Method: An experiment using a mockup consisting of a HUD and a driver's seat was conducted with 28 participants. A passenger car, motorcycle, and crossing pedestrian were presented as visual objects on a 65-in monitor behind the image of the HUD. Participants were asked questions under different conditions with changes in the HUD luminance and asked to correctly answer questions regarding changes in the inter-vehicle distance and the existence of visual objects on the road. The number of correct answers provided by the participants was recorded.

Results: The experimental results revealed that the HUD had an effect under the twilight condition wherein the background luminance was low. Furthermore, analyzing the number of correct answers for each visual target revealed that the HUD had an effect under conditions wherein the size of the visual object was small.

Conclusion: It was experimentally derived that the luminance ratio $R_{\text{luminance}} = 1.1$ was the threshold value that affected the recognition of the visual object in the background when the HUD was presented in front of the driver's field of view.

Implication: The threshold value of the luminance ratio $R_{\text{luminance}}$ obtained in this study can be considered when deciding how to properly display the HUD in front of the driver's field of view.

1. Introduction

Head-up displays (HUDs) are increasingly being installed in automobiles. The idea of a HUD is that superimposes an imaginary image of an instrument on an image of the outside world was first proposed in the 1940s. It was not until 1960 that HUDs were produced for use in military aircrafts, while now are widely used in commercial aircrafts as well. In automobiles, they began to be used in the mid-1980s, and are now used in a wide range of vehicles [1]. Since around 1990, the effects of HUDs have been compared with those of head-down displays [2,3]. Compared to conventional instrument panels, HUDs have the advantage of making

it easier for drivers to check road conditions and maintain lane control [4].

Current HUDs are placed below the driver's field of view and present information on navigation routes, vehicle speeds, and road signs. The evolution of augmented reality (AR) technology is expected to further popularize them. By applying AR, information can be obtained by overlaying the HUD on the vehicle in front, pedestrian crossing, or display along the curbline [5,6]. Furthermore, the entire windshield can be used as a display area for HUD [7]. A study comparing the effectiveness of AR in visual and auditory modalities, not only for providing information, but also for presenting warnings, has shown the

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effectiveness and acceptability of visual AR displays [8]. Studies investigating pedestrian crossings and road signs have also shown that many drivers do not recognize pedestrian crossings and drive above the maximum speed limit of the signs, and that it is possible to intervene from the infrastructure side by making pedestrian crossings and signs more visually prominent [9]. Such visual enhancement can be achieved not only from the infrastructure side, such as painting the road, but also by AR-HUD, and enhancement by AR-HUD is expected to be effective. In addition, AR-HUD with a sense of depth can increase the driver's attention span and thus reduce the workload compared with normal HUD displays [10]. Therefore, AR-HUDs offer several advantages. Because it is presented in front of the driver's field of view more than a conventional HUD, it is expected to be more effective in presenting information.

However, there are concerns that this may cause visual obstruction. For example, when a HUD display is filled with navigational information, the background of the display may not be sufficiently transparent to allow another car on the road to be clearly seen [11]. If a motorcycle is behind a filled HUD image, the driver may not see it until it is too late. One study investigated the preferred position of HUD in an actual driving environment; many experimental participants preferred the HUD in the visual field below the line of sight and outside the traffic scene [12]. For these reasons, an AR-HUD must be optimized such that it is both effective as a display and safe to use without creating any visual obstructions to the driver's field of view [13].

Currently, UN Regulation No. 125, the international regulation for the forward field of vision of automobile drivers, defines a HUD that is presented in front of the driver as a field of view assistant (FVA), rather than a conventional HUD, and discusses its requirements. There are situations in which quantitative indicators are required for FVA.

The purpose of this study is to investigate the effect of the HUD on the forward field of view of the driver. To this end, we focused on HUD luminance and conducted an experimental study on a display luminance that could achieve both proper visibility and safety for drivers. A bench test using a HUD was conducted in the darkroom of the Japan Automobile Research Institute (JARI). A mockup consisting of the HUD and driver's seat in Fig. 1 was constructed and set up to observe a 65-in monitor located 5.0 m from the forward field of view of the driver. An actual traffic scene in the background was projected on this monitor and observed through a HUD image adjusted to constant luminance; the effect of HUD luminance on the ability of the driver to recognize the traffic scene in the background was then investigated. Fig. 2 shows the flowchart of the analysis in this study. First, we compared daytime and twilight conditions. Next, by analyzing the number of correct answers for each visual object, we derived the upper limit threshold for HUD luminance.

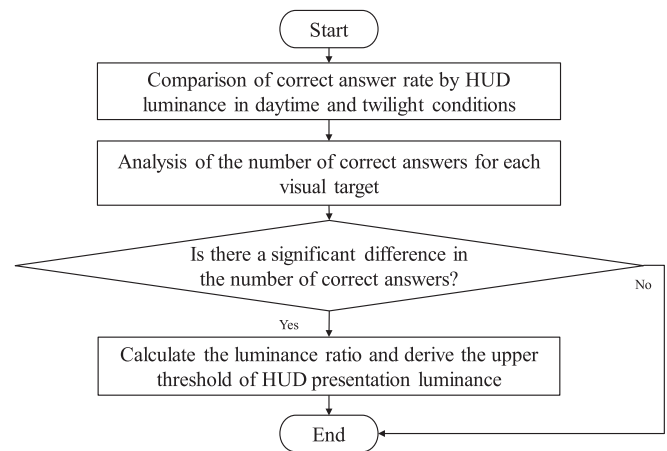


Fig. 2. Flowchart of the analysis.

2. Experiment participants and apparatus used

2.1. Participants

There were 28 participants, 14 men and 14 women, in the experiment. The men were aged 21–66 (mean 42.1 years, SD 14.5 years) and the women were aged 21–66 (mean 42.9 years, SD 14.3). To ensure uniformity in the age distribution of the participants, we selected three people in their 20s to 50s and two people in their 60s belonging to both genders. All participants had a driver's license and drove daily. Further, they were recruited from the general public, were not engaged in the research and development of automobiles or HUDs, and had no experience using HUDs on a daily basis. This study was approved by the Experimental Ethics Committee of the Japan Automobile Research Institute (JARI) and informed consent was obtained from all participants.

The purpose of the experiment was explained to the participants using an instruction sheet, and they were asked to confirm their characteristics, such as age and gender, their driving experience, and whether they had previously used an HUD. Before starting the experimental task, a practice task was conducted using scenes not used in the experiment. This ensured that the participants understood the task before performing it.

2.2. Apparatus

The traffic scene in the background was displayed on a 65-in 4 K resolution liquid crystal display (TCL 65C601U), as shown in Fig. 1. The distance between the center of the display and the ocular point was set to 5.0 m. For the HUD, a projector unit with a digital micromirror device

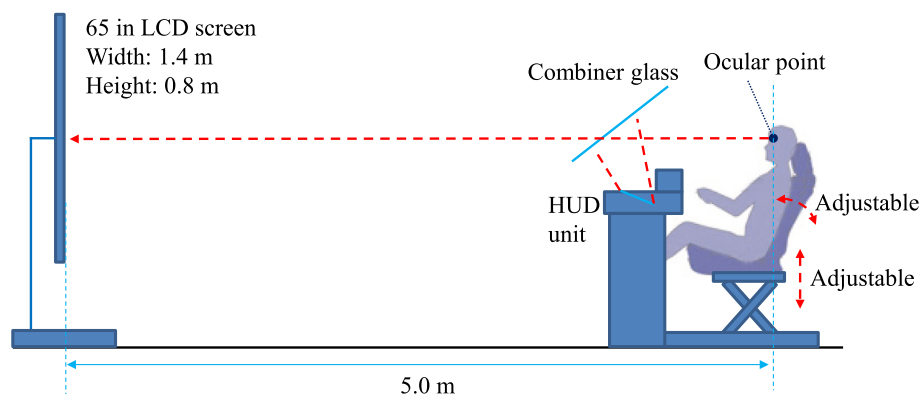


Fig. 1. HUD mockup.

(DMD) with element resolution of 854×480 px was used, while the display content was controlled using a laptop PC (Lenovo ThinkPad X250). The seat was positioned such that the height and back inclination could be adjusted, allowing participants to see forward from the same visual position. The vertical viewing angle of the white part of the combiner glass in the HUD was approximately 5.9° near the center. A color luminance meter (Topcon BM-5AS) was used to measure luminance.

3. Procedure

To evaluate the effect of HUDs on the forward visibility of a person, we evaluated the extent to which participants could comprehend the traffic situation in front of a vehicle when given a very short time to observe it. For this purpose, traffic scene images were presented for a short period of time and the ability of the participants to discriminate between them was examined. We included three types of visual objects in a traffic scene: passenger cars, motorcycles, and crossing pedestrians. When the passenger car and motorcycle were the visual targets, we asked the participants to answer questions regarding changes in the distance between the two still images before and after HUD luminance decreased. When a pedestrian was the visual target, we asked for answers regarding the presence or absence of crossing pedestrians and the crossing direction in one still image. Fig. 3 shows the time and flow for presenting still images of passenger cars, motorcycles, and crossing pedestrians.

Regarding still images presented for a short period, if they were presented for too long, the difference in driving decisions may have been too large, and it would have been easier to identify the differences from reality. However, if the presentation time was too short, it would have been difficult to fully comprehend what was presented. In a study that analyzed the gaze behavior of drivers under three types of traffic situations, the average gazing time of drivers on the road ahead ranged from 0.7 to 1.5 s [14]. In another study, the average gazing times of novice and expert drivers involved in visual searches were approximately 0.3 and 0.5 s, respectively [15]. Another study that analyzed the gazing behavior of drivers showed that the average gazing time of drivers for visual information in front of them on the road varied, ranging from 0.24 to 1.03 s [16].

From these studies, we inferred that the minimum time for a driver to recognize visual information on the road ahead would be approximately 0.3 s. Below, we discuss the evaluation items and presentation methods, followed by different visual targets.

(1) Visual target: Passenger car or motorcycle

Evaluation items: Change in inter-vehicle distance.

Presentation method:

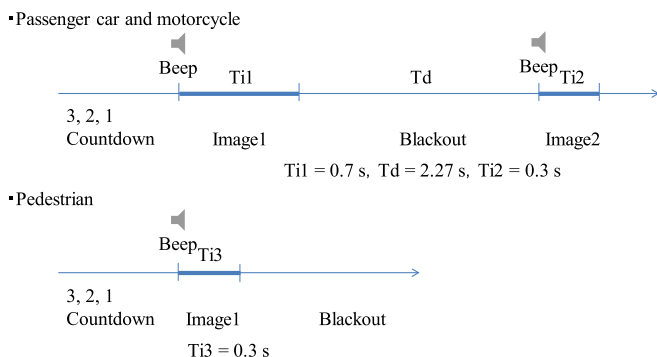


Fig. 3. Time and flow for image presentation.

- After counting from three to one, two still images were displayed in succession with a blackout interval, accompanied by a beep signal.
- There were two possible changes in the inter-vehicle distance: it was either increased or decreased by 10 m.
- The presentation times of the first and second images were set to 0.7 and 0.3 s, respectively.
- The blackout interval was set because the time when the inter-vehicle distance changed due to a deceleration of 0.4 g (3.92 m/s^2) in 10 m was 2.27 s. The deceleration of 0.4 g was set because it is often used as a threshold for strong braking [17–19].

(2) Visual target: Pedestrian

Evaluation items: Existence of pedestrian and crossing direction.

Presentation method:

- After counting from three to one, one still image was presented along with a beep signal.
- There were three possible conditions for the pedestrians: they either crossed in a direction from the right or left hand of the vehicle or did not cross at all.
- The image presentation time was set to 0.3 s.

For passenger cars and motorcycles, the initial inter-vehicle initial distances were set at 30 and 50 m, respectively, with or without a change in distance (10 m shortened (deceleration), unchanged, or 10 m increased (acceleration)). Three types of pedestrians were selected: those who crossed from the left, those who crossed from the right, or those who did not cross at all. In addition to daylight conditions, wherein the visual targets were easier to distinguish, we also set a twilight condition, wherein the brightness of the images was reduced. We focused on whether the experimental participants noticed the sudden deceleration of the passenger car or motorcycle in front of the vehicle. Accordingly, the change in inter-vehicle distance was based on two patterns: no change or short change. However, if there were only two patterns, it would be easy for participants to predict them; therefore, we added some conditions to increase the inter-vehicle distance and make the prediction more challenging.

Visibility through HUD may also depend on the content displayed. Currently, HUDs commonly display content such as speed and navigation arrows, but in the future, they may be used to highlight objects on the road. All these can distract the driver's attention but are unlikely to significantly obstruct their field of view. However, if the entire display area of the HUD were to light up brightly owing to a malfunction or for some other reason, it could obstruct an extremely wide field of view compared to speed and navigation displays. Assuming this worst-case scenario, this study examined the brightness requirements that would allow background objects to be recognized when the HUD is displayed over such a wide field of view. If the requirements for visibility in this worst-case scenario are clarified, we believe that visibility will be ensured, even if a large area is rendered by AR-HUDs in the future. Therefore, we filled the entire HUD with a single white color and set the display luminance from 0 to 500 cd/m^2 in increments of 100 cd/m^2 . The virtual image distance of the HUD was set to 5 m, which was equidistant from the LCD, and the HUD was presented to overlap the visual object.

We evaluated the number of correct answers that participants provided under 156 conditions. The combinations of these conditions are shown in Table 1 and examples of the visual targets are shown in Figs. 4–6.

4. Results

4.1. Comparison of daylight and twilight conditions

We examined the effect of HUD luminance on the visual perception of the participants based on the rate of correct answers provided under

Table 1

Experimental conditions.

Brightness	Object	Initial distance	Option	HUD luminance [cd/m ²]
Daylight	Passenger car	50 m	Unchanged	0
			Shorten	(No image, baseline condition)
			Increased	
	Motorcycle	30 m	Unchanged	100
			Shorten	200
			Increased	300
Twilight	Passenger car	50 m	Unchanged	400
			Shorten	500
			Increased	
		30 m	Unchanged	
			Shorten	
			Increased	
	Crossing pedestrian	50 m	No pedestrian	
			Crossing from the left	
			Crossing from the right	
		30 m	Unchanged	
			Shorten	
			Increased	
	Motorcycle	50 m	Unchanged	
			Shorten	
			Increased	
		30 m	Unchanged	
			Shorten	
			Increased	
	Crossing pedestrian	50 m	No pedestrian	
			Crossing from the left	
			Crossing from the right	

different conditions, including changes in inter-vehicle distance and crossing direction. Two-way analysis of variance (ANOVA) was performed on the brightness of daylight and twilight conditions and HUD luminance. The interaction between the differences in HUD luminance and brightness under daylight and twilight conditions was significant ($F = 8.6561$, $p < 0.001$). The main effects of the differences in HUD luminance ($F = 13.8734$, $p < 0.001$) and brightness ($F = 45.3325$, $p <$

0.001) were also significant. Fig. 7 shows the average correct answer rates. Subsequently, because both the interaction and main effects were significant, a simple main effects test was performed. A significant difference was observed between daylight and twilight conditions when the HUD luminance was 200 cd/m² or more. A significant difference was observed only between the HUD luminance values under twilight conditions. No such significant difference was found between these values under the daylight conditions. Fig. 8 presents the results of multiple comparisons using the Bonferroni method under twilight conditions.

4.2. Analysis of the number of correct answers for each visual target

A two-way ANOVA confirmed the effects of HUD luminance under twilight conditions. Therefore, we analyzed the differences in the number of correct answers for each visual target (passenger car, motorcycle, and pedestrian) under twilight conditions. Chi-square analysis was used to analyze the relationship between the HUD luminance and the number of correct/false answers for each of the 13 twilight conditions. The following three conditions showed significant relationships in the chi-square analysis ($p < 0.001$): twilight/motorcycle/50 m/increased, twilight/pedestrian/50 m/crossing from the right, and twilight/pedestrian/50 m/crossing from the left. Figs. 9–11 show the number of correct answers under these three conditions; the HUD luminance condition for which the p -value of the adjusted standardized residuals was significant as a result of the residual analysis and the subject's visual perception when the HUD luminance was 500 cd/m², the most difficult viewing condition.

Under low-light conditions, the effect of HUD was not observed when a passenger car was the visual target or when the distance to the motorcycle remained the same or decreased. On the other hand, the three conditions under which there were significant differences due to the HUD were those in which the distance to the motorcycle increased and those in which the pedestrians crossed the street. This indicates that the HUD had an effect when the size of the visual object was small and luminance was low.

5. Discussion

The experimental results revealed that the HUD was effective under

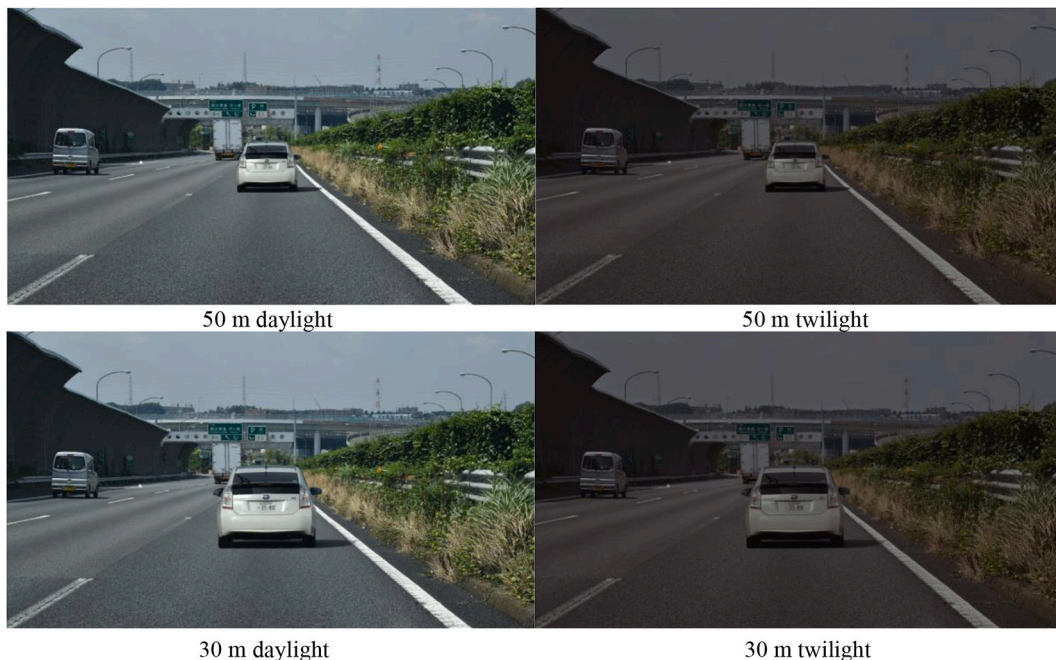


Fig. 4. Passenger car under different conditions.



Fig. 5. Motorcycle under different conditions.

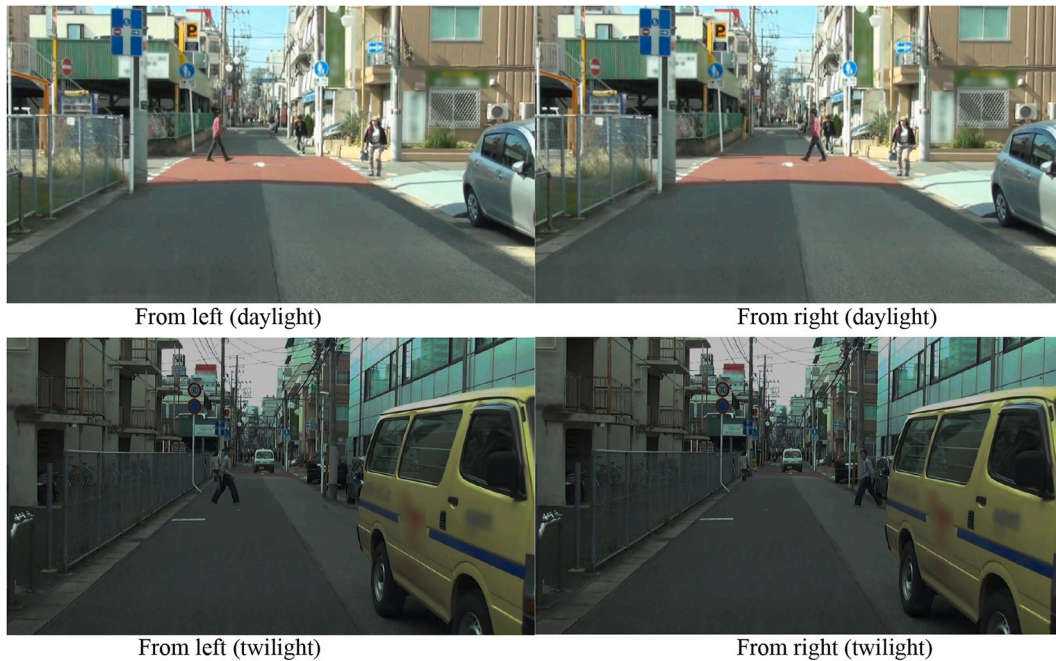


Fig. 6. Pedestrian under different conditions.

twilight conditions, wherein background brightness was low. Additionally, the analysis of the number of correct answers for each visual target revealed three conditions with significant differences. We considered the effects of HUD luminance under these three conditions. Fig. 12 illustrates the concept of luminance used in this study. The luminance ratio $R_{\text{luminance}}$ is defined based on the concept of the contrast ratio [20], as shown in Eq. (1):

$$R_{\text{luminance}} = (L_{\text{object}} + L_{\text{HUD}}) / (L_{\text{background}} + L_{\text{HUD}}) \quad (1)$$

where L_{object} is the luminance of the visual object (motorcycle/pedestrian), L_{HUD} is the luminance of the virtual image on the HUD combiner,

and $L_{\text{background}}$ is the luminance of the background, which is the road surface in this case.

Fig. 13 shows the results for the luminance measurements of the target and background under the condition of “twilight/ motorcycle/50 m/increased.” The luminance of the largest taillight was considered the luminance of the visual target. Figs. 14 and 15 show the results of the luminance measurements of the target and background under the “twilight/pedestrian/50 m/crossing from right” and “twilight/pedestrian/50 m/crossing from left” conditions, respectively. Under these two conditions, we adopted the largest luminance of the upper body as the luminance of the crossing pedestrians.

Fig. 16 shows the $R_{\text{luminance}}$ values derived for each HUD luminance.

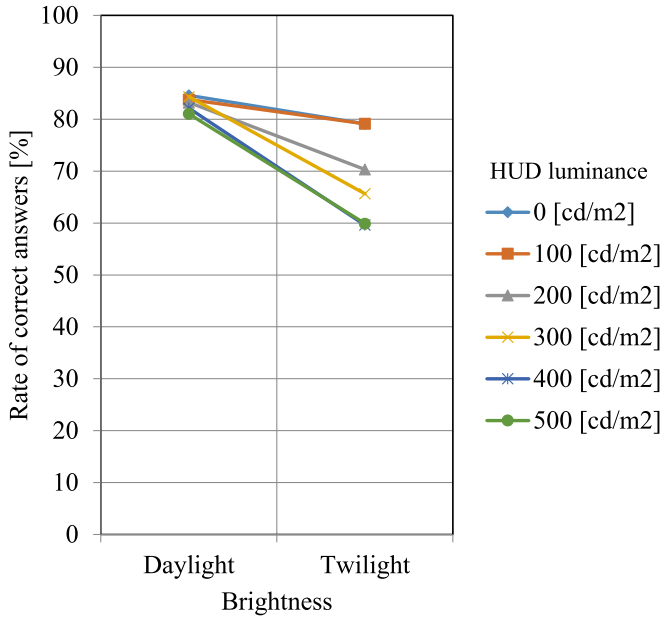


Fig. 7. Mean value of the percentage of correct answers.

From the residual analysis in the “twilight/motorcycle/50 m/increased” condition in Fig. 9, the number of correct answers was significantly higher for no HUD and a HUD luminance of 100 cd/m² and significantly lower for a HUD luminance of 400 cd/m² or higher. This suggests that a HUD luminance of 200 cd/m² or more corresponds to an $R_{\text{luminance}}$ value of 1.08 or less and that the HUD presentation may reduce the percentage of correct answers. The number of correct answers decreased significantly when the HUD luminance was 400 cd/m² or more and the $R_{\text{luminance}}$ was 1.04 or less. Similarly, in the residual analysis for the “twilight/pedestrian/50 m/crossing from right” condition in Fig. 10, the number of correct answers was significantly higher for no HUD and a HUD luminance of 100 cd/m² and significantly lower for a HUD luminance of 400 cd/m² or higher. This suggests that the HUD luminance of 200 cd/m² or more corresponds to an $R_{\text{luminance}}$ value of 1.04 or less and that the HUD presentation may reduce the percentage of correct answers. The number of correct answers decreased significantly when the

HUD luminance was 400 cd/m² or higher and $R_{\text{luminance}}$ was 1.02 or lower. However, in the residual analysis for the “twilight/pedestrian/50 m/crossing from left” condition in Fig. 11, the number of correct answers was significantly higher for the conditions of no HUD and HUD luminance of 100 cd/m² conditions and significantly lower for the HUD luminance of 300 cd/m² or higher. This suggests that a HUD luminance of 200 cd/m² or more corresponds to an $R_{\text{luminance}}$ value of 1.04 or less and that the HUD presentation may reduce the percentage of correct answers. The number of correct answers decreased significantly when the HUD luminance was 300 cd/m² or more and $R_{\text{luminance}}$ was 1.03 or lower.

Based on the above results, an $R_{\text{luminance}}$ of approximately 1.08 was close to the boundary of the change in the number of correct answers. Considering the safety margin, we set $R_{\text{luminance}} = 1.1$ as the threshold value and controlled HUD luminance such that $R_{\text{luminance}}$ would be 1.1 or less, to ensure the visibility of objects in front. Based on Eq. (1), Eq. (2) can be obtained by setting $R_{\text{luminance}}$ to less than 1.1 and solving for HUD luminance.

$$L_{\text{HUD}} = 10^* (L_{\text{Object}} - 1.1 * L_{\text{Background}}) \quad (2)$$

By using Eq. (2), we determined the threshold of HUD luminance to avoid affecting the recognition of visual targets by the driver when the HUD was presented in front of the driver's field of view.

An NHTSA report [1] shows the results of a study of HUD luminance using Eq. (3) below, which considers only the contrast of HUD luminance with respect to background luminance because the HUDs used so far are small and placed only below the frontal area of the visual field.

$$R_{\text{contrast}} = (L_{\text{HUD}} + L_{\text{Background}}) / L_{\text{Background}} \quad (3)$$

Solving Eq. (3) for L_{HUD} yields Eq. (4):

$$L_{\text{HUD}} = (R_{\text{contrast}} - 1) * L_{\text{Background}} \quad (4)$$

The lower limit of contrast is given by $R_{\text{contrast}} = 1.15$ [1], which, when substituted into Eq. (4), yields Eq. (5).

$$L_{\text{HUD}} = 0.15 * L_{\text{Background}} \quad (5)$$

However, the results of the study on the upper luminance limit have not been presented at a sufficient contrast ratio because it is difficult for conventional HUDs to produce high luminance. The results of this study show the upper luminance limit when a HUD is presented in front of the field of view; it is thought that an appropriate luminance range can be

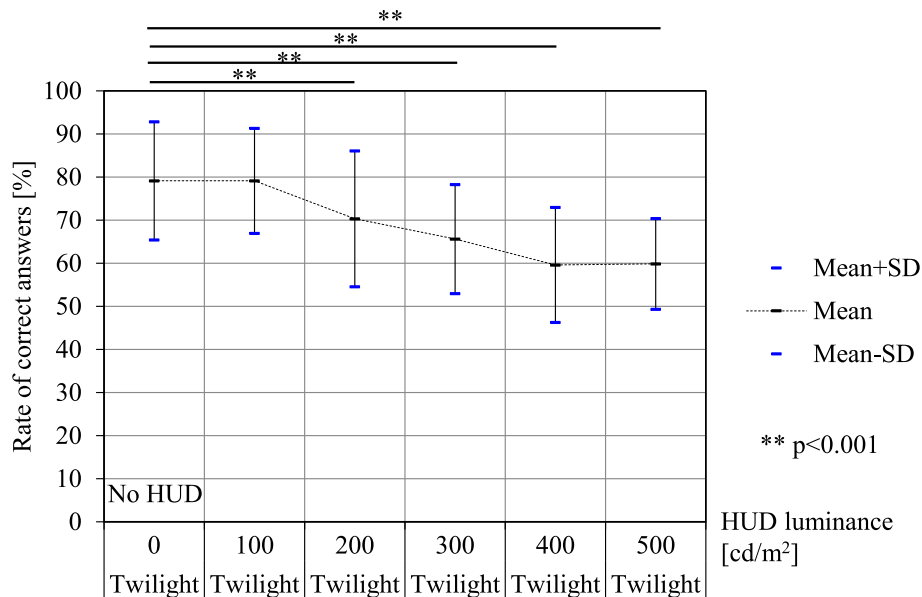
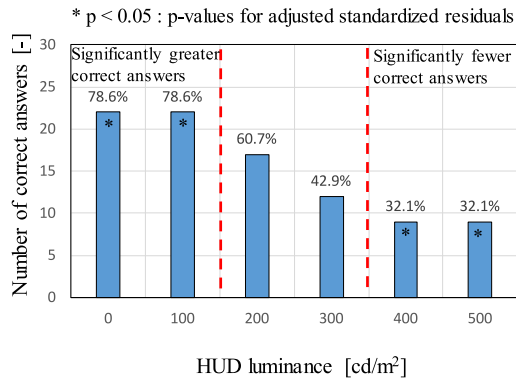
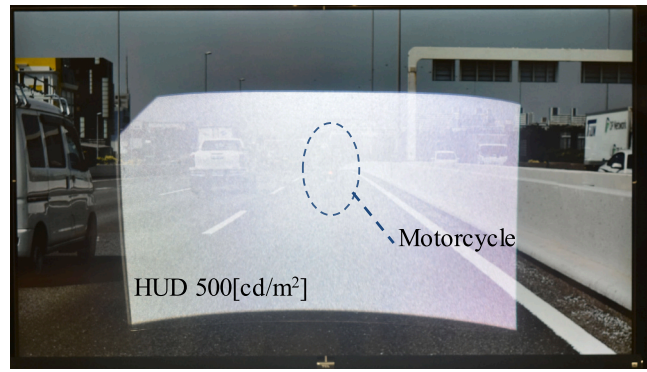


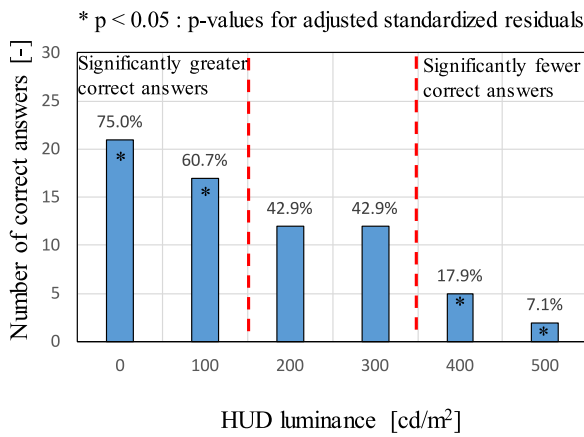
Fig. 8. Variation in the rate of correct answers under twilight conditions due to HUD luminance.



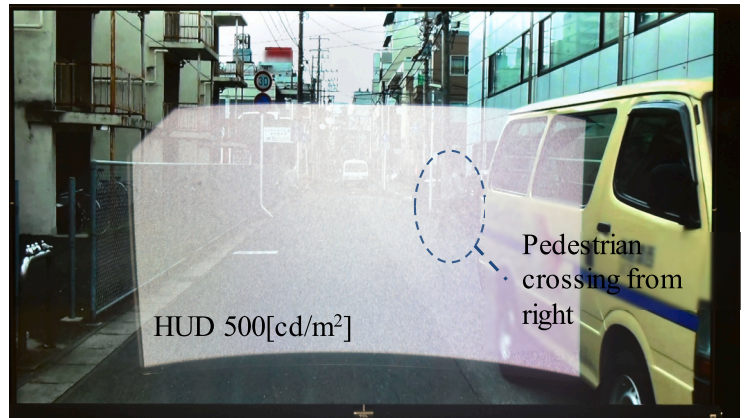
(a) Number of correct answers



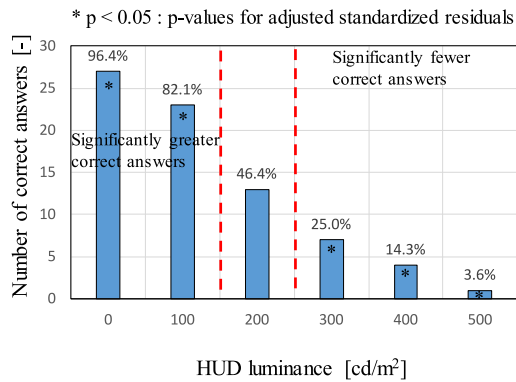
(b) Image from ocular point

Fig. 9. Results under the “twilight/motorcycle/50 m/increased” condition.

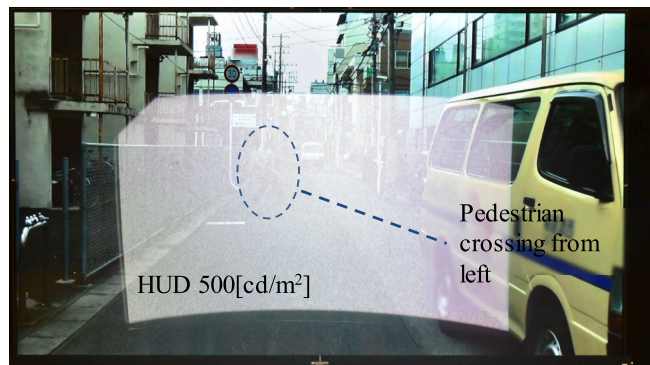
(a) Number of correct answers



(b) Image from ocular point

Fig. 10. Results under the “twilight/pedestrian/50 m/crossing from right” condition.

(a) Number of correct answers



(b) Image from ocular point

Fig. 11. Results under the “twilight/pedestrian/50 m/crossing from left” condition.

obtained by combining it with the lower luminance limit of the conventional study. In addition, the latest HUDs are brighter than conventional HUDs, with figures up to $30,000 \text{ cd/m}^2$ as of 2015 [21]. The use of LEDs and lasers as light sources and improvements in the reflectivity of HUD projection surfaces will further increase the brightness of HUDs in the future, and meaning need for the upper limit value shown in this study is expected to increase.

We attempt to derive the maximum and minimum thresholds for

HUD luminance by applying Eqs. (2) and (5) to actual cases. Table 2 shows examples of the measurement results of the luminance of a white passenger car and the road surface in a real environment, as shown in Fig. 17, as well as the HUD luminance thresholds derived from these results. Assuming that the white passenger car was the visual target, the upper limit of HUD luminance was approximately 400 cd/m^2 after sunset. Under daylight conditions, HUD luminance did not affect target detection in our experiments. The threshold value derived using Eq. (2)

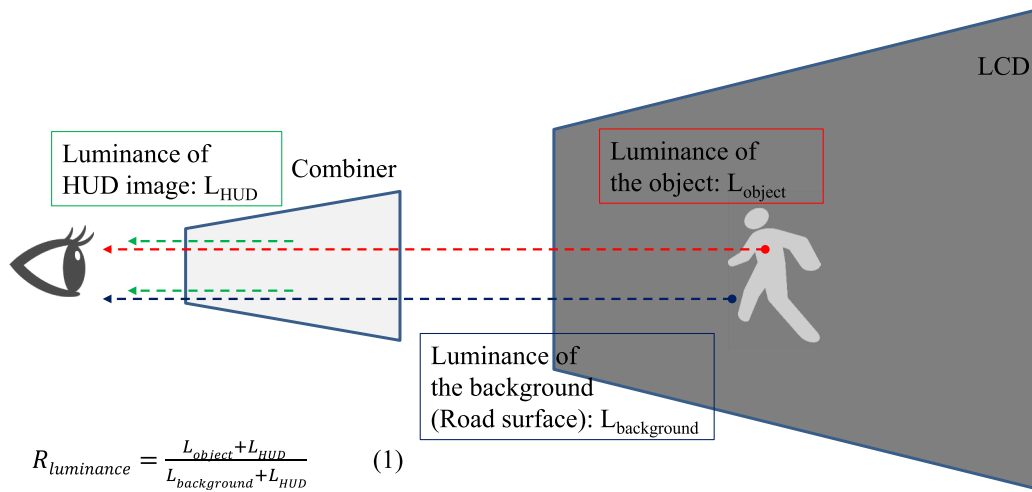
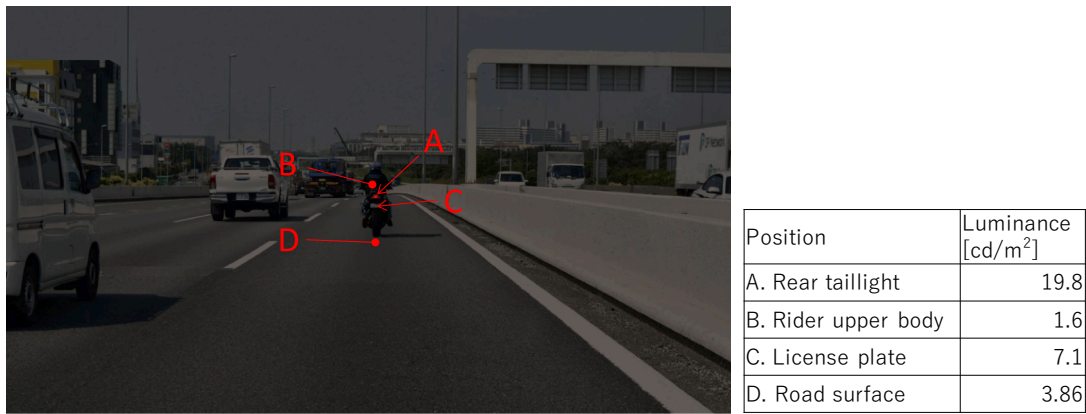
Fig. 12. Concept of luminance ratio $R_{luminance}$.

Fig. 13. Results of the luminance measurement under the “twilight/motorcycle/50 m/increased” condition.

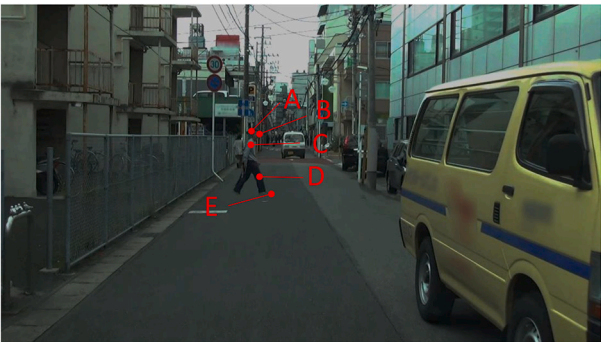


Fig. 14. Results of the luminance measurement under the “twilight/pedestrian/50 m/crossing from right” condition.

is sufficiently large compared with the luminance of HUDs currently available in the market. Therefore, the key considerations are setting the luminance of the visual targets and road surface in situations where the ambient brightness is low, such as twilight. For example, not only passenger cars, but also motorcycles and pedestrians can be considered visual objects, and the luminance of the road surface may vary depending on the type of pavement. A study by Bozorg et al. [22], who investigated the illumination of road surfaces and pedestrians by headlamps, road lighting, etc. at night, showed that the average luminance of pedestrians was 11.6 cd/m² and that of road surfaces was 0.46 cd/m² when road lighting was 50%, the pedestrian target was 50 m

away, and the road surface was dry. By substituting these values into Eq. (2) to derive the upper limit of HUD luminance, we obtained a result of 111 cd/m². This is approximately one-fourth the upper limit for passenger cars after sunset, as listed in Table 2. Therefore, the luminance of the viewed object and background varied depending on the actual road surface and lighting conditions. Future studies should investigate luminance under various conditions and derive the upper and lower limits based on the conditions assumed.

Regarding the number of participants in an experiment, for example, in a questionnaire survey such as [23], several hundred or more participants were surveyed to ensure data reliability. However, in an



Position	Luminance [cd/m ²]
A. Pedestrian's hair	6.30
B. Pedestrian's face	9.31
C. Pedestrian's upper body	20.4
D. Pedestrian's leg	2.57
E. Road surface	12.3

Fig. 15. Results of the luminance measurement under the “twilight/pedestrian/50 m/crossing from left” condition.

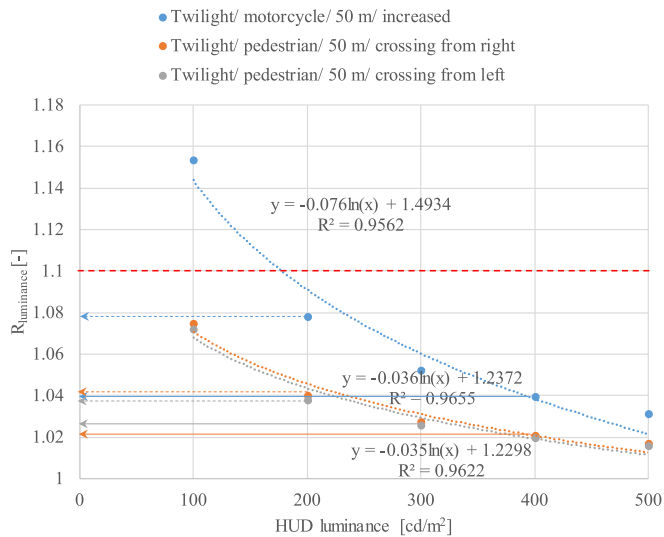


Fig. 16. Calculation results of $R_{luminance}$ using HUD luminance.

Table 2
Outdoor luminance measurement results and derived HUD luminance thresholds.

Luminance[cd/m ²]	Outdoor daylight		Outdoor twilight	
	Sunny	Lightly cloudy	Before sunset	After sunset
White passenger car	31,100	8020	516	53
Road surface (asphalt)	3030	899	77.7	9.64
Maximum L_{HUD} (Using Eq. (2))	277,670	70,311	4305	424
Minimum L_{HUD} (Using Eq. (5), if the background is the white passenger car)	4665	1203	77.4	7.95
Minimum L_{HUD} (Using Eq. (5), if the background is the road surface)	454.5	134.85	11.66	1.45

ergonomic experiment such as this one, it is generally difficult to conduct a study with several hundred participants owing to budget constraints. The larger the number of participants in an experiment is, the more accurate the data are; however, there is much debate as to how many participants should be considered. We referred to the NHTSA Guidelines for Visual Hand Control Distraction [24] for an ergonomic experiment of the same type. The guidelines specify that the number of participants in the experiment should be 24 (12 men and 12 women), with three male and three female participants in each of the four age groups: 18–24, 25–39, 40–54, and 55 years or older. Although there was a slight difference in the way the age groups were delimited in this study,



Fig. 17. Scene for luminance measurement.

the number of participants and the male-to-female ratio were almost the same. In addition, the number of participants in the studies that conducted evaluation experiments using HUDs with experimental participants was 30 for [25], 24 for [26], 30 for [27], 10 for [28], and 30 for [29]. The number of participants in this study (28) is similar in similar experiments. The small number of participants in the experiment is a limitation of this study and we believe that the statistical treatment ensures the validity of the results to some extent; however, it is possible that unobserved heterogeneity remains: all experimental participants were Japanese, that is, all had brown irises. It is known that differences in iris color have a significant effect on intraocular stray light, especially in low-light conditions, which affects glare sensitivity [30]. As such effects were not examined in this study, future validation experiments with a larger number of participants are required.

6. Conclusions

This study examined the luminance effects of a HUD on the presence, relative change in distance, and crossing direction of visual objects, including a passenger car, motorcycle, and pedestrian, in front of the driver in a traffic environment presented on a 65-in. LCD monitor for a short time. The HUD image, which was white on all sides, was largely presented in the field of view in front of the driver to avoid the driver being affected by the presented content. The luminance varied from 0 cd/m² (“no HUD” condition) to 500 cd/m² in increments of 100 cd/m².

A comparison between daylight and twilight conditions showed that the rate of correct answers provided by the participants was significantly lower under twilight conditions. Under daylight conditions, even with the highest HUD luminance of 500 cd/m², there was no significant difference in the movement of the visual target or in the rate of correct answers for detection. However, under twilight conditions, significant

differences in the aforementioned factors were observed when HUD luminance was 200 cd/m² or higher.

Under twilight conditions, we compared the number of correct answers for each visual target, finding significant differences in the conditions of motorcycles and pedestrians 50 m away. We derived the luminance ratios for the three conditions in which there was a significant difference and examined the threshold value.

By using Eq. (2), we determined the threshold of HUD luminance to avoid affecting the recognition of visual targets when the HUD was presented in front of the driver's field of view. The threshold value of L_{HUD} was calculated by substituting the luminance of the real environment into Eq. (2). This threshold value was sufficiently high under daylight conditions, indicating that the effect of HUD was limited. However, when the luminance of the visual target was low, such as after sunset, the effect of HUD luminance has to be considered.

This study led to the development of a HUD luminance threshold that affects the discrimination of visual objects seen in front of the driver when the HUD is presented in front of the driver's field of vision. When applying this threshold to a real environment, the problem of defining the luminance values of these objects and that of the background surface (the road) must be considered. Currently, there is no definition of standard luminance for visible objects such as passenger cars, motorcycles, and pedestrians on roads or a standard luminance for road surfaces. In the future, these luminances should be defined and the influence of luminance under low-light conditions, such as twilight, should be investigated.

CRedit authorship contribution statement

Takashi Hosokawa: Writing – original draft, Formal analysis, Data curation. **Hiroshi Hashimoto:** Project administration. **Takayuki Tsui:** Resources, Conceptualization. **Keisuke Saito:** Writing – review & editing. **Tomotaka Igarashi:** Writing – review & editing, Project administration. **Akinari Hirao:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors report no competing interests.

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References

- [1] L.W. Gish, L. Staplin, *Human Factors Aspects of Using Head up Displays in Automobiles: A Review of the Literature*, DOT HS 808 320, NHTSA, 1995.
- [2] S. Okabayashi, M. Furukawa, M. Sakata, T. Hatada, Driver's ability to recognize the forward view and head-up display image in automobiles, *J. Light. Vis. Environ.* 16–2 (1992) 61–69, <https://doi.org/10.2150/jlve.16.2.13>.
- [3] M. Smith, J. Streeter, G. Burnett, J.L. Gabbard, Visual search tasks: the effects of head-up displays on driving and task performance, in: *Proceedings 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 2015, pp. 80–87, <https://doi.org/10.1145/2799250.2799291>.
- [4] H. Ito, B. Atsumi, H. Uno, M. Akamatsu, A.R. DriveSim, Visual distraction while driving: trends in research and standardization, *IATSS Res.* 25 (2001) 20–28, [https://doi.org/10.1016/S0386-1112\(14\)60067-4](https://doi.org/10.1016/S0386-1112(14)60067-4).
- [5] L. Abdi, F. Ben Abdallah, A. Meddeb, In-vehicle augmented reality traffic information system: a new type of communication between driver and vehicle, *Proc. Comp. Sci.* 73 (2015) 242–249, <https://doi.org/10.1016/j.procs.2015.12.024>.
- [6] J. Park, W. Park, Functional requirements of automotive head-up displays: a systematic review of literature from 1994 to present, *Appl. Ergon.* 76 (2019) 130–146, <https://doi.org/10.1016/j.apergo.2018.12.017>.
- [7] W. Wu, F. Blaicher, J. Yang, T. Seder, D. Cui, A prototype of landmark-based car navigation using a full-windshield head-up display system, in: *Proceedings of the 2009 Workshop on Ambient Media Computing (ACM)*, 2009, pp. 21–28, <https://doi.org/10.1145/1631005.1631012>.
- [8] F. Schwarz, W. Fastenmeier, Augmented reality warnings in vehicles: effects of modality and specificity on effectiveness, *Accid. Anal. Prev.* 101 (2017) 55–66, <https://doi.org/10.1016/j.aap.2017.01.019>.
- [9] E.M. Acerra, C. Lantieri, V. Vignali, M. Pazzini, S. Andrea, Safety roads: the analysis of driving behaviour and the effects on the infrastructural design, *Transport. Res. Proc.* 69 (2023) 336–343, <https://doi.org/10.1016/j.trpro.2023.02.180>.
- [10] R. Li, Y.V. Chen, L. Zhang, Z. Shen, Z.C. Qian, Effects of perception of head-up display on the driving safety of experienced and inexperienced drivers, *Displays* 64 (2020) 1–10, <https://doi.org/10.1016/j.displa.2020.101962>.
- [11] J.L. Gabbard, M. Smith, K. Tanous, H. Kim, B. Jonas, A.R. DriveSim, An immersive driving simulator for augmented reality head-up display research, *Front. Robot. AI* 6 (2019) 98, <https://doi.org/10.3389/frobt.2019.00098>.
- [12] P. Tretten, A. Gärling, R. Nilsson, T.C. Larsson, An on-road study of head-up display: preferred location and acceptance levels, in: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 55, 2011, pp. 1914–1918, <https://doi.org/10.1177/1071181311551398>.
- [13] A. Pauzie, Head up display in automotive: a new reality for the driver, in: *International Conference of Design, User Experience, and Usability*, 2015, pp. 505–516, https://doi.org/10.1007/978-3-319-20889-3_47.
- [14] X. Li, A. Vaezipour, A. Rakotonirainy, S. Demmler, Effects of an in-vehicle eco-safe driving system on driver's glance behaviour, *Accid. Anal. Prev.* 122 (2019) 143–152, <https://doi.org/10.1016/j.aap.2018.10.007>.
- [15] P. Chapman, G. Underwood, K. Roberts, Visual search patterns in trained and untrained novice drivers, *Transp. Res. F* 5 (2002) 157–167, [https://doi.org/10.1016/S1369-8478\(02\)00014-1](https://doi.org/10.1016/S1369-8478(02)00014-1).
- [16] M. Sodhi, B. Reimer, I. Llamazares, Glance analysis of driver eye movements to evaluate distraction, *Behav. Res. Methods Instrum. Comput.* 34 (2002) 529–538, <https://doi.org/10.3758/BF03195482>.
- [17] G. Howe, G. Xu, D. Hoover, D. Elsasser, F. Barickman, *Commercial connected-vehicle test procedure development and test results—emergency electronic brake light* (Report No. DOT HS 812 327), 2016, p. 109.
- [18] S. McClellan, *Smart Cities in Application: Healthcare, Policy, and Innovation*, Springer, 2019, <https://doi.org/10.1007/978-3-030-19396-6>.
- [19] Y. Xue, S. Chihuri, H.F. Andrews, M.E. Betz, C. DiGiuseppi, D.W. Eby, L.L. Hill, V. Jones, T.J. Mielenz, L.J. Molnar, D. Strogatz, B.H. Lang, T. Kelley-Baker, G. Li, Potentially inappropriate medication use and hard braking events in older drivers, *Geriatrics* 6 (2021) 20, <https://doi.org/10.3390/geriatrics6010020>.
- [20] R.D. Guenther, D.G. Steel, L.P. Bayvel, *Encyclopedia of Modern Optics*, “DISPLAYS, R. L. Donofrio”, 2005, pp. 366–376, <https://doi.org/10.1016/B012-369395-0/01255-0>.
- [21] A. Rankin, J. Thompson, Next-generation head-up displays, *Inform. Display* 31 (2015) 18–21, <https://doi.org/10.1002/j.2637-496X.2015.tb00810.x>.
- [22] S. Bozorg, E. Tetri, I. Kosonen, T. Luttinen, The effect of dimmed road lighting and car headlights on visibility in varying road surface conditions, *LEUKOS* 14 (2018) 259–273, <https://doi.org/10.1080/15502724.2018.1452152>.
- [23] T.U. Saeed, M.W. Burris, S. Labi, K.C. Sinha, An empirical discourse on forecasting the use of autonomous vehicles using consumers' preferences, *Technol. Forecast. Soc. Change* 158 (2020) 120130, <https://doi.org/10.1016/j.techfore.2020.120130>.
- [24] *National Highway Traffic Safety Administration, Guidelines for Reducing Visual-Manual Driver Distraction during Interactions with Integrated, In-Vehicle, Electronic Devices*, Docket No. NHTSA-2010-0053, 2013.
- [25] K. Stojmenova Pečecnik, S. Tomazič, J. Sodnik, Design of head-up display interfaces for automated vehicles, *Int. J. Hum.-Comput. Stud.* 177 (2023) 103060, <https://doi.org/10.1016/j.ijhcs.2023.103060>.
- [26] C. Maag, N. Schomig, F. Naujoks, I. Karl, A. Keinath, A. Neukum, Measuring workload effects of augmented reality head-up displays using detection response task, *Transp. Res. F* 92 (2022) 201–219, <https://doi.org/10.1016/j.trf.2022.11.010>.
- [27] S. Wang, Z. Li, Y. Wang, Q. Zheng, Z. Liu, Effectiveness of connected-vehicle-based curve speed warning systems with considerations of heads-up display language type's effects on driver behavior, *Transp. Res. F* 99 (2023) 127–148, <https://doi.org/10.1016/j.trf.2023.10.009>.
- [28] A.W.D. Bremers, A. Özgür Yöntem, K. Li, D. Chu, V. Meijering, C.P. Janssen, Perception of perspective in augmented reality head-up displays, *Int. J. Hum.-Comput. Stud.* 155 (2021) 102693, <https://doi.org/10.1016/j.ijhcs.2021.102693>.
- [29] L. Pfannmüllera, M. Kramer, B. Senner, K. Bengler, A comparison of display concepts for a navigation system in an automotive contact analog head-up display, *Proc. Manuf.* 3 (2015) 2722–2729, <https://doi.org/10.1016/j.promfg.2015.07.678>.
- [30] C. Nischler, R. Michael, C. Wintersteller, P. Marvan, L.J. van Rijn, J.E. Coppens, T. J. van den Berg, M. Emesz, G. Grabner, Iris color and visual functions, *Graefes Arch. Clin. Exp. Ophthalmol.* 1 (2013) 195–202, <https://doi.org/10.1007/s00417-012-2006-8>.